# Solving linear systems

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Consider a linear algebra problem with n linear equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots\cdots\cdots\cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases} \tag{1}$$

which can be denoted by

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \tag{2}$$

where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \ \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}. \tag{3}$$

**Remark 0.1.** In the present age of AI, solving a system of linear equations becomes more and more important.

As we all known that for this problem, when $\det \mathbf{A} \neq 0$, there is an unique solution by the Cramer's rule, that is

$$x_i = \frac{\det \mathbf{A}_i}{\det \mathbf{A}} \ (i = 1, 2, \cdots, n). \tag{4}$$

Thus the cost of this solution is $(n+1)(n-1)n!$. For example, when $n = 30$, this cost is around $2.38 \times 10^{35}$. Impossible to be calculated by computer, when $n$ is larger.

# 1 Gaussian elimination

Recall that three useful operations can be applied to a linear system of equations that yield an equivalent system, meaning that one has the same solutions. These operations are as follows:

**(1)** Swap one equation for another.

**(2)** Add or subtract a multiple of one equation from another.

**(3)** Multiply an equation by a nonzero constant.

Because of these, we can solve a system of $n$ linear equations in $n$ unknowns consists of a set of algebraic relations of the form

$$\sum_{j=1}^{n} a_{ij}x_j = b_i,\ i = 1, \cdots, n \tag{5}$$

where $x_j$ are the unknowns, $a_{ij}$ are the coefficients of the system and $b_i$ are the components of the right hand side.

In particular, we solve a nonsingular $3 \times 3$ system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3. \end{aligned} \tag{6}$$

System (6) can be more conveniently written in so-called tableau form:

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

There are two steps of Gaussian elimination of solving the above problem.

**The elimination step:** To carry out the elimination step, we need to put zeros in the lower triangle, using the allowed row operations.

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22} & a_{23} & b_2 \\ 0 & 0 & a_{33} & b_3 \end{array}\right]$$

We can write the elimination step as the loop

```
for j = 1 : 2
  eliminate column j
end
```

where, by "eliminate column $j$," we mean "use row operations to put a zero in each location below the main diagonal." For example, to carry out elimination on column 1, we need to put zeros in $a_{21}, a_{31}$. This can be written as the following loop within the former loop:

```
for j = 1 : 2
  for i = j+1 : 3
    eliminate entry a(i,j)
  end
end
```

It remains to fill in the inner step of the double loop, to apply a row operation that sets the $a_{ij}$ entry to zero. For example, the first entry to be eliminated is the $a_{21}$ entry. To accomplish this, we subtract $\frac{a_{21}}{a_{11}}$ times row 1 from row 2, assuming that $a_{11} \neq 0$. That is, the first two rows change from

$$\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \end{array}$$

to

$$\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} & b_2 - \frac{a_{21}}{a_{11}}b_1 \end{array}$$

Accounting for the operations, this requires one division (to find the multiplier $a_{21}/a_{11}$), plus 3 multiplications and 3 additions.

Inserting this step into the same double loop results in

```
for j = 1 : 2
  if abs(a(j,j))<eps; error(zero pivot encountered); end
  for i = j+1 : 3
    mult = a(i,j)/a(j,j);
    for k = j+1:3
      a(i,k) = a(i,k) - mult*a(j,k);
    end
    b(i) = b(i) - mult*b(j);
  end
end
```

We can make a total count of operations for the elimination step of Gaussian elimination.

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 2(3)+1 & 0 & 0 & 0 \\ 2(3)+1 & 2(2)+1 & 0 & 0 \end{array}\right]$$

**For general $n \times n$ system:**

**Code:**

```
for j = 1 : n-1
  if abs(a(j,j))<eps; error(zero pivot encountered); end
  for i = j+1 : n
    mult = a(i,j)/a(j,j);
    for k = j+1:n
      a(i,k) = a(i,k) - mult*a(j,k);
    end
    b(i) = b(i) - mult*b(j);
  end
end
```

**Operations:**

$$\left[\begin{array}{ccccccc|c} 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 2n+1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 2n+1 & 2(n-1)+1 & 0 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & & & 0 \\ \vdots & \vdots & \vdots & & & & & 0 \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \cdots & 0 & 0 & 0 & 0 \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \cdots & 2(3)+1 & 0 & 0 & 0 \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \cdots & 2(3)+1 & 2(2)+1 & 0 & 0 \end{array}\right]$$

**Count the total operations:**

**Lemma 1.1.** For any positive integer $n$,

**(a)** $1+2+3+4+\cdots+n=\frac{n(n+1)}{2}$,

**(b)** $1^2+2^2+3^2+4^2+\cdots+n^2=\frac{n(n+1)(2n+1)}{6}$.

It is convenient to add up the operations in reverse order of how they are applied. Starting on the right, we total up the operations as

$$\begin{aligned}\sum_{j=1}^{n-1}\sum_{i=1}^{j}[2(j+1)+1] &= \sum_{j=1}^{n-1}[2j(j+1)+j] \\ &= 2\sum_{j=1}^{n-1}j^2+3\sum_{j=1}^{n-1}j = 2\frac{(n-1)n(2n-1)}{6}+3\frac{(n-1)n}{2} \\ &= (n-1)n\left[\frac{2n-1}{3}+\frac{3}{2}\right]=\frac{n(n-1)(4n+7)}{6} \\ &= \frac{2}{3}n^3+\frac{1}{2}n^2-\frac{7}{6}n,\end{aligned}$$

where Lemma 1.1 has been applied.

**The back-substitution step:** To do a back-substitution

$$\begin{aligned}x_3 &= \frac{b_3}{a_{33}} \\ x_2 &= \frac{b_2-a_{23}x_3}{a_{22}} \\ x_1 &= \frac{b_1-a_{12}x_2-a_{13}x_3}{a_{11}}\end{aligned}$$

**For general $n\times n$ system,** we carry out the following back-substitution step,

$$\begin{aligned}x_n &= \frac{b_n}{a_{nn}} \\ x_{n-1} &= \frac{b_{n-1}-a_{n-1,n}x_n}{a_{n-1,n-1}} \\ &\vdots \\ x_2 &= \frac{b_2-a_{23}x_3-\cdots-a_{2n}x_n}{a_{11}} \\ x_1 &= \frac{b_1-a_{12}x_2-a_{13}x_3-\cdots-a_{1n}x_n}{a_{11}}\end{aligned}$$

Because of the triangular shape of the nonzero coefficients of the equations, we start at the bottom and work our way up to the top equation. In this way, the required $x_i$'s are known when they are needed to compute the next one. Counting operations yields

$$1 + 3 + 5 + \cdots + (2n-1) = \sum_{i=1}^{n} (2i-1) = 2\sum_{i=1}^{n} i - \sum_{i=1}^{n} 1 = 2\frac{n(n+1)}{2} - n = n^2. \tag{7}$$

In MATLAB syntax, the back-substitution step is

```
for i = n : -1 : 1
  for j = i+1 : n
    b(i) = b(i) - a(i,j)*x(j);
  end
  x(i) = b(i)/a(i,i);
end
```

Check the $3 \times 3$ system!

# 2 LU factorization

Let's study the LU factorization by solving an example below. Several questions are to be answered.

**1.** How to find the lower triangular $L$ and the upper triangular $U$, such that $A = LU$?

**2.** How many operations of the LU factorization?

**3.** Why do we study "how" of the LU factorization?

**Example 2.1.** Solve the following system by (a) finding the LU factorization, (b) checking $LU = A$ and then (c) carrying out the two-step back substitution.

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -6 \end{bmatrix}$$

**Solution of (a):**
Here are elimination steps for getting the upper triangular $U$.

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} \longrightarrow \begin{array}{c}\text{subtract} \quad 2\times\text{row1} \\ \text{from row2}\end{array} \longrightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ -3 & 1 & 1 \end{bmatrix}$$

$$\longrightarrow \begin{array}{c}\text{subtract} -3\times\text{row1} \\ \text{from row3}\end{array} \longrightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 7 & -2 \end{bmatrix}$$

$$\longrightarrow \begin{array}{c}\text{subtract} -\frac{7}{3}\times\text{row2} \\ \text{from row3}\end{array} \longrightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = U.$$

The lower triangular $L$ matrix is formed, by putting 1's on the main diagonal and the multipliers in the lower triangular—in the specific places they were used for elimination. That is,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix}.$$

**Solution of (b):**
Check that

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} = A.$$

**Solution of (c):**
The $Lc = b$ step is

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -6 \end{bmatrix},$$

which corresponds to the system

$$\begin{aligned} c_1 &= 3 \\ 2c_1 + c_2 &= 3 \\ -3c_1 - \frac{7}{3}c_2 + c_3 &= -6. \end{aligned}$$

Starting at the top, the solutions are $c_1 = 3, c_2 = -3, c_3 = -4$.

The $Ux = c$ step is

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ -4 \end{bmatrix},$$

which corresponds to the system

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 3 \\ -3x_2 &= -3 \\ -2x_3 &= -4 \end{aligned}$$

and is solved from the bottom up to give $x = [3, 1, 2]$.

**Answer to question 1.** *How to find the lower upper triangular $L$ and the upper triangular $U$, such that $A = LU$?*

The above procedure shows us that:

**To find** $U$ We apply several row operations to eliminate the entries in the lower triangular, which is the same procedure as Gaussian elimination we did. Then we have $U$.

**To find** $L$ The lower triangular $L$ matrix is formed, by putting 1's on the main diagonal and the multipliers in the lower triangular—in the specific places they were used for elimination.

**A new question arises:** Why can we set $L$ by the above procedure?

**For the above example** You can check that $LU = A$.

**For the general case** The reason that this procedure gives the LU factorization follows from three facts about lower triangular matrices.

**FACT 1** Let $L_{ij}(-c)$ denote the **lower triangular matrix** whose only nonzero entries are 1's on the main diagonal and $-c$ in the (i,j) position. Then $A \longrightarrow L_{ij}(-c)A$ represents the row operation "subtracting $c$ times row $j$ from row $i$". For example, to eliminate the lower entry $a_{21}$,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \longrightarrow \begin{array}{c} \text{subtract } \frac{a_{21}}{a_{11}} \times \text{row1} \\ \text{from row2} \end{array} \longrightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - \frac{a_{21}}{a_{11}}a_{11} & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

is equivalent to

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ -\frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = L_{21}\left(-\frac{a_{21}}{a_{11}}\right) A$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

**FACT 2** $L_{ij}(-c)^{-1} = L_{ij}(c)$.

For example,

$$\begin{bmatrix} 1 & 0 & 0 \\ -\frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

since

$$\begin{bmatrix} 1 & 0 & 0 \\ -\frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

**FACT 3** The following matrix product equation holds.

$$\begin{bmatrix} 1 & 0 & 0 \\ c_1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c_2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c_3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ c_1 & 1 & 0 \\ c_2 & c_3 & 1 \end{bmatrix}.$$

Go back to Example 2.1, the elimination steps is equivalent to

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{7}{3} & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = U. \text{ (based on FACT 1)}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{7}{3} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \text{ (based on FACT 2)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = LU \text{ (based on FACT 3)}$$

**Answer to question 2.** *How many operations of the LU factorization?*

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ -\frac{a_{21}}{a_{11}} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

The above row operation require one division plus 2 multiplications and 2 subtractions. For eliminating $a_{32}$, it require one division plus 1 multiplications and 1 subtractions. The following matrix presents the operations for eliminating the lower entries.

$$\begin{bmatrix} 0 & 0 & 0 \\ 2(2)+1 & 0 & 0 \\ 2(2)+1 & 2(1)+1 & 0 \end{bmatrix}.$$

For general matrix $A = (a_{ij})_{n\times n}$, we can make a total count of operations for the elimination steps:

$$\begin{bmatrix} 0 & & & & & \\ 2(n-1)+1 & 0 & & & & \\ 2(n-1)+1 & 2(n-2)+1 & 0 & & & \\ \vdots & \vdots & \ddots & \ddots & & \\ 2(n-1)+1 & 2(n-2)+1 & \cdots & 2(2)+1 & 0 & \\ 2(n-1)+1 & 2(n-2)+1 & \cdots & 2(2)+1 & 2(1)+1 & 0 \end{bmatrix}.$$

Starting on the right, we total up the operations as

$$\begin{aligned} \sum_{j=1}^{n-1}\sum_{i=1}^{j}[2(j)+1] &= \sum_{j=1}^{n-1} j[2(j)+1] \\ &= 2\sum_{j=1}^{n-1} 2j^2 + \sum_{j=1}^{n-1} j \\ &= 2\frac{(n-1)n(2n-1)}{6} + \frac{(n-1)n}{2} \\ &= \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \end{aligned}$$

where we apply

**(a)** $1 + 2 + 3 + \cdots + n = n(n+1)/2$,

**(b)** $1^2 + 2^2 + 3^2 + \cdots + n^2 = n(n+1)(2n+1)/6$.

So the elimination step for getting $L$ and $U$ can be completed in $\frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n.$ operations.

Note that the forward-substitution step and the back substitution step is cheaper in operations. For example, counting operations of the back substitution step, which is

$$\begin{aligned} x_1 &= \frac{b_1 - a_{12}x_2 - \cdots - a_{1n}x_n}{a_{11}} \\ x_2 &= \frac{b_2 - a_{23}x_3 - \cdots - a_{2n}x_n}{a_{22}} \\ &\vdots \\ x_n &= \frac{b_n}{a_{nn}} \end{aligned}$$

yields

$$1 + 3 + 5 + \cdots + (2n-1) = \sum_{i=1}^{n} 2i - 1 = 2\sum_{i=1}^{n} i - \sum_{i=1}^{n} 1 = 2\frac{n(n+1)}{2} - n = n^2.$$

**Answer to question 3.** *Why do we study "how" of the LU factorization?*

For solving a set of problems

$$\begin{aligned} Ax &= b_1 \\ Ax &= b_2 \\ &\vdots \\ Ax &= b_k \end{aligned}$$

with various right-hand side vector $b_i$.

Classical Gaussian elimination will require approximately $2kn^3/3$ operations, where $A$ is an $n \times n$ matrix, since we must start over at the beginning for each problem.

For the LU approach, the elimination is only done for $Ax = b_1$, followed by two back substitution ($Lc = b, Ux = c$) for each $b_i$. Therefore, the approximate number of operations with the LU approach is $2n^3/3 + 2kn^2$.

### 2.4.1 Gaussian elimination with partial pivoting

Recall that we have a question left to be addressed.

**Question 1.** Why do we call Gaussian elimination as "**naive**" Gaussian elimination?

**Counter-example 1.** We have mentioned an example like

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Gaussian elimination fails to solve this example because $a_{11} = 0$.

**Counter-example 2.** Sometimes $a_{11} \neq 0$ and $a_{11}$ is a very small number.

**Example 2.2.** For example, consider

$$\begin{aligned} 10^{-20}x_1 + x_2 &= 1, \\ x_1 + 2x_2 &= 4. \end{aligned}$$

We will solve this equations 3 times:

1. with complete accuracy.
2. in double precision Matlab by Gaussian elimination.
3. exchanged the order of the equations first and then solved in double precision Matlab by Gaussian elimination.

Solutions:

1. **with complete accuracy.**
   In tableau form, Gaussian elimination proceeds as

$$\begin{bmatrix} 10^{-20} & 1 & | & 1 \\ 1 & 2 & | & 4 \end{bmatrix} \longrightarrow \begin{array}{c} \text{subtract} \quad 10^{20} \times \text{row1} \\ \text{from row2} \end{array} \longrightarrow \begin{bmatrix} 10^{-20} & 1 & | & 1 \\ 0 & 2 - 10^{20} & | & 4 - 10^{20} \end{bmatrix}$$

   The bottom equation is

$$(2 - 10^{20})x_2 = 4 - 10^{20} \longrightarrow x_2 = \frac{4 - 10^{20}}{2 - 10^{20}}$$

   and the top equation yields

$$\begin{aligned} 10^{-20}x_1 + \frac{4 - 10^{20}}{2 - 10^{20}} &= 1 \\ x_1 = 10^{20}\left(1 - \frac{4 - 10^{20}}{2 - 10^{20}}\right) & \\ x_1 = \frac{-2 \times 10^{20}}{2 - 10^{20}}. & \end{aligned}$$

   The exact solution is

$$[x_1, x_2] = \left[\frac{-2 \times 10^{20}}{2 - 10^{20}}, \frac{4 - 10^{20}}{2 - 10^{20}}\right] \approx [2, 1].$$

2. **in double precision Matlab by Gaussian elimination.**
   The computer version of Gaussion elimination proceeds slightly differently:

$$\begin{bmatrix} 10^{-20} & 1 & | & 1 \\ 1 & 2 & | & 4 \end{bmatrix} \longrightarrow \begin{array}{c} \text{subtract} \quad 10^{20} \times \text{row1} \\ \text{from row2} \end{array} \longrightarrow \begin{bmatrix} 10^{-20} & 1 & | & 1 \\ 0 & 2 - 10^{20} & | & 4 - 10^{20} \end{bmatrix}$$

In double precision MATLAB, $2 - 10^{20}$ is the same as $-10^{20}$, due to rounding. Similarly, $4 - 10^{20}$ is stored as $-10^{20}$. Now the bottom equation is

$$-10^{20}x_2 = -10^{20} \longrightarrow x_2 = 1.$$

The machine arithmetic version of the top equation becomes

$$10^{-20}x_1 + 1 = 1,$$

so $x_1 = 0$. The computed solution is exactly

$$[x_1, x_2] = [0, 1].$$

This solution has large relative error compared with the exact solution.

3. **exchanged the order of the equations first and then solved in double precision Matlab by Gaussian elimination.**
We repeat the computer version of Gaussian elimination, after changing the order of the two equations:

$$\left[\begin{array}{ccc} 1 & 2 & 4 \\ 10^{-20} & 1 & 1 \end{array}\right] \longrightarrow \begin{array}{c} \text{subtract } 10^{-20} \times \text{row1} \\ \text{from row2} \end{array} \longrightarrow \left[\begin{array}{ccc} 1 & 2 & 4 \\ 0 & 1 - 2 \times 10^{-20} & 1 - 4 \times 10^{-20} \end{array}\right]$$

In double precision MATLAB, $1 - 2 \times 10^{-20}$ is stored as 1 and $1 - 4 \times 10^{-20}$ is stored as 1. The equations are now

$$\begin{aligned} x_1 + 2x_2 &= 4 \\ x_2 &= 1, \end{aligned}$$

which yield the computed solution $x_1 = 2$ and $x_2 = 1$. Of course, this is not the exact answer, but it is correct up to approximately 16 digits.

**Answer to Question 1.** Now we know the reason why Gaussian elimination is "naive". The reason is that there are two serious difficulties of Gaussian elimination:

1. encountering a zero pivot
2. encountering a very small pivot like example 2.4

A question arises:

**Question 2.** How shall we avoid such a situation?

Recall that the procedure to eliminate $a_{21}$

$$\left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array}\right] \longrightarrow \begin{array}{c} \text{subtract } \frac{a_{21}}{a_{11}} \times \text{row1} \\ \text{from row2} \end{array} \longrightarrow \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} - \frac{a_{21}}{a_{11}}a_{12} & a_{23} - \frac{a_{21}}{a_{11}}a_{13} \\ a_{31} & a_{32} & a_{33} \end{array}\right] \left|\begin{array}{c} b_1 \\ b_2 - \frac{a_{21}}{a_{11}}b_1 \\ b_3 \end{array}\right.$$

So there are two difficulties when $a_{11} = 0$ or $a_{11}$ is a very small number.

Recall that doing the **swap row** operation, the solutions of the system is unchanged.

**Answer to Question 2.** For solving

$$\left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array}\right] \left|\begin{array}{c} b_1 \\ b_2 \\ b_3 \end{array}\right.,$$

we do the following procedure to eliminate $a_{21}, a_{31}, a_{32}$.

To eliminate $a_{21}, a_{31}$,

**Step 1.** To locate the largest entry of the first column and we swap its row with the pivot row.

- e.g., If $|a_{21}|$ is the largest entry among $|a_{11}|, |a_{21}|, |a_{31}|$, we swap row 2 with row 1.

**Step 2.** To do Gaussian elimination for eliminating the new $a_{21}, a_{31}$.

To eliminate $a_{32}$,

**Step 1.** To locate the largest entry between $|a_{22}|$ and $|a_{32}|$.

- e.g., If $|a_{32}|$ is the largest entry between $|a_{22}|$ and $|a_{32}|$, we swap row 3 with row 2.

**Step 2.** To do Gaussian elimination for eliminating the new $a_{32}$.

To do an example,

**Example 2.3.** Apply Gaussian elimination with partial pivoting to solve the system

$$\begin{aligned} x_1 - x_2 + 3x_3 &= -3 \\ -x_1 - 2x_3 &= 1 \\ 2x_1 + 2x_2 + 4x_3 &= 0. \end{aligned}$$

This system is written in tableau form as

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & -3 \\ -1 & 0 & -2 & 1 \\ 2 & 2 & 4 & 0 \end{array}\right].$$

Under partial pivoting we compare $|a_{11}| = 1$ with $|a_{21}| = 1$ and $|a_{31}| = 2$, and choose $a_{31}$ for the new pivot. This is achieved through an exchange of row 1 and 3:

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & -3 \\ -1 & 0 & -2 & 1 \\ 2 & 2 & 4 & 0 \end{array}\right] \longrightarrow \begin{array}{c}\text{exchange row1} \\ \text{from row 3}\end{array} \longrightarrow \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ -1 & 0 & -2 & 1 \\ 1 & -1 & 3 & -3 \end{array}\right]$$

$$\longrightarrow \begin{array}{c}\text{subtract } -\frac{1}{2} \times \text{row1} \\ \text{from row 2}\end{array} \longrightarrow \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & -1 & 3 & -3 \end{array}\right]$$

$$\longrightarrow \begin{array}{c}\text{subtract} \frac{1}{2} \times \text{row1} \\ \text{from row3}\end{array} \longrightarrow \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & 1 & -3 \end{array}\right]$$

Before eliminating column 2 we must compare the current $|a_{22}|$ with the current $|a_{32}|$. Because the latter is larger, we again switch rows:

$$\left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & 1 & -3 \end{array}\right] \longrightarrow \begin{array}{c}\text{exchange row2} \\ \text{from row 3}\end{array} \longrightarrow \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & -2 & 1 & -3 \\ 0 & 1 & 0 & 1 \end{array}\right]$$

$$\longrightarrow \begin{array}{c}\text{subtract } -\frac{1}{2} \times \text{row2} \\ \text{from row 3}\end{array} \longrightarrow \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & -2 & 1 & -3 \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} \end{array}\right].$$

Now that all three multipliers are less than 1 in absolute value.

The equations are now simple to solve. From

$$\begin{aligned} \frac{1}{2}x_3 &= -\frac{1}{2} \\ -2x_2 + x_3 &= -3 \\ 2x_1 + 2x_2 + 4x_3 &= 0, \end{aligned}$$

we find that $x = [1, 1, -1]$.

Homework: Apply Gaussian elimination with partial pivoting to solve the system

$$\begin{bmatrix} 3 & 1 & 2 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$$

## 2.5 Iterative methods

Until now, we can solve

$$Ax = b$$

by the following **direct methods**:

1. naive Gaussian elimination,
2. LU approach,
3. Gaussian elimination with pivoting.

- Recall that the complexity of all of above three approaches is of order $n^3$.
- If we try to solve $A_{1000\times1000}x_{1000\times1} = b_{1000\times1}$, it needs 1 billion operations.

So for solving linear systems of small dimension, we can use the above direct methods. For solving linear systems of large dimension, what shall we do?

In order to solve linear systems of large dimension, we can use the iterative method to solve it.

So let's start a new topic, 2.5 Iterative methods.

**Recall that** we have discussed the idea of Fixed-Point Iteration.

1. To find the solution $r$, such that $f(r) = 0$.
2. We rewrite $f(x) = 0$ as $x = g(x)$.
3. We do the iteration
$$x_{i+1} = g(x_i),$$
starting with an initial guess.

★ **Convergence theorem:** Assume that $g$ is continuously differentiable, that $g(r) = r$, the that $S = |g'(r)| < 1$. Then Fixed-Point Iteration converges linearly with rate $S$ to the fixed point $r$ for initial guesses.

**That is,** Fixed-Point Iteration is locally convergent if $|g'(r)| < 1$.

**Now** iterative methods also can be applied to solving systems of linear equations.

1. To find the solution $x$, such that Ax-b=0. We will solve it approximately.
2. We rewrite $Ax - b = 0$ as
$$x = G(x).$$
3. We do the iteration
$$x^{(i+1)} = G(x^{(i)}),$$
starting with an initial guess.

★ **Convergence theorem ???**

**2.5.1 Jacobi Method** So let's study the first iterative approach, 2.5.1 **Jacobi method**.

Let's first write the system of equations $Ax = b$ and its detailed form

$$\sum_{j=1}^{n} a_{ij}x_j = b_i,\ 1 \le i \le n.$$

In order to rewrite $Ax = b$ into a form of $x = G(x)$, we should keep something to the left hand side and move everything else to the right hand side: So we keep $x_i$ in $i$-th equation to left hand side,

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{ii}\boxed{x_i} + \cdots + a_{in}x_n = b_i,\ 1 \le i \le n.$$

and move everything else to the right hand side. Then we have

$$x_i = \left(b_i - \sum_{j=1, j\neq i}^{n} a_{ij}x_j\right)/a_{ii}.$$

It means we obtain $\vec{x} = G(\vec{x})$ with some linear function $G$. If $\vec{x}$ is the solution of $Ax - b = 0$, $\vec{x}$ is the fixed point. Here we assume that all diagonal entries are nonzero. If this is not the case, we can usually rearrange the equation so that it is.

Let's do the Jacobi Iteration:

**Step 1.** Choose initial guess $x^{(0)} = (x_1^{(0)}, x_2^{(0)}, \cdots, x_n^{(0)})^T$.

**Step 2.** We do the following iterations

**for** $k = 0, 1, 2, \cdots, K-1$, where $K$ is the number of iterations:

**for** $i = 1, 2, \cdots, n$

$$x_i^{(k+1)} = \left(b_i - \sum_{j=1, j\neq i}^{n} a_{ij}x_j^{(k)}\right)/a_{ii}.$$

**end**

**end**

We can choose any initial guess. We always choose $x^{(0)} = (0, 0, \cdots, 0)^T$.

For example,

**Example 2.4.** Apply the Jacobi Method to the system

$$3x_1 + x_2 = 5,$$
$$x_1 + 2x_2 = 5.$$

Begin by solving the first equation for $x_1$ and the second equation for $x_2$. We will use the initial guess $[x_1^{(0)}, x_2^{(0)}]^T = [0, 0]^T$. We have

$$x_1 = \frac{5 - x_2}{3}$$
$$x_2 = \frac{5 - x_1}{2}.$$

In fact we have the exact solution $[1, 2]^T$. So we can compare the exact solution and the numerical solutions below.

The two equations are iterated:

$$\begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} |0-1| \\ |0-2| \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(0)}}{3} \\ \frac{5-x_1^{(0)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-0}{3} \\ \frac{5-0}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{5}{2} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{5}{3}-1\right| \\ \left|\frac{5}{2}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(1)}}{3} \\ \frac{5-x_1^{(1)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-5/2}{3} \\ \frac{5-5/3}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{6} \\ \frac{5}{3} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{5}{6}-1\right| \\ \left|\frac{5}{3}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{3} \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(3)} \\ x_2^{(3)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(2)}}{3} \\ \frac{5-x_1^{(2)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-5/3}{3} \\ \frac{5-5/6}{2} \end{bmatrix} = \begin{bmatrix} \frac{10}{9} \\ \frac{25}{12} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{10}{9}-1\right| \\ \left|\frac{25}{12}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{1}{9} \\ \frac{1}{12} \end{bmatrix}$$

Further steps of Jacobi show convergence toward the exact solution.

**2.5.2 Gauss-Seidal Method** Closely related to the Jacobi Method is an iteration called the Gauss-Seidal Method. Note that Gauss-Seidal often **converges faster** than Jacobi if the method is convergent.

In the Jacobi method, the equations are solved in order of $i = 1, 2, \cdots, n$,

$$x_i^{(k+1)} = \left( b_i - \sum_{j=1, j\neq i}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}.$$

For $i$-th equation, we have already got $x_j^{(k+1)}, j = 1, \cdots, i-1$. We clearly know that $x_j^{(k+1)}$ is more accurate than $x_j^{(k)}, j = 1, \cdots, i-1$. So taking into account of this, we change the Jacobi Method

$$x_i^{(k+1)} = \left( b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}$$

to Gauss-Seidal Method

$$x_i^{(k+1)} = \left( b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}.$$

Returning to Example 2.4, we see that Gauss-Seidal looks like this: The two equations are iterated:

$$\begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} |0-1| \\ |0-2| \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(0)}}{3} \\ \frac{5-x_1^{(1)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-0}{3} \\ \frac{5-5/3}{2} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{5}{3} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{5}{3}-1\right| \\ \left|\frac{5}{3}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(1)}}{3} \\ \frac{5-x_1^{(2)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-5/3}{3} \\ \frac{5-10/9}{2} \end{bmatrix} = \begin{bmatrix} \frac{10}{9} \\ \frac{35}{18} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{10}{9}-1\right| \\ \left|\frac{35}{18}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{1}{9} \\ \frac{1}{18} \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(3)} \\ x_2^{(3)} \end{bmatrix} = \begin{bmatrix} \frac{5-x_2^{(2)}}{3} \\ \frac{5-x_1^{(3)}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5-35/18}{3} \\ \frac{5-55/54}{2} \end{bmatrix} = \begin{bmatrix} \frac{55}{54} \\ \frac{215}{108} \end{bmatrix}, \qquad \begin{bmatrix} \left|\frac{55}{54}-1\right| \\ \left|\frac{215}{108}-2\right| \end{bmatrix} = \begin{bmatrix} \frac{1}{54} \\ \frac{1}{108} \end{bmatrix}$$

We can clearly see that Gauss-Seidal **converges faster** than Jacobi

Now we have the following question:

★ **Convergence theorem ???** Does Jacobi or Gauss-Seidal convergence for any problem?

**Answer:** No!

**Counter-example:** We swap the equations in Example 2.4.

$$x_1 + 2x_2 = 5,$$
$$3x_1 + x_2 = 5.$$

Below we apply Jacobi to solve. Solve the first equation for $x_1$ and the second equation for $x_2$. We will use the initial guess $[x_1^{(0)}, x_2^{(0)}]^T = [0, 0]^T$. We have

$$x_1 = 5 - 2x_2$$
$$x_2 = 5 - 3x_1.$$

In fact we have the exact solution $[1, 2]^T$. So we can compare the exact solution and the numerical solutions below.

The two equations are iterated as before, but the result are quite different:

$$\begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} |0-1| \\ |0-2| \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$
$$\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = \begin{bmatrix} 5 - 2x_2^{(0)} \\ 5 - 3x_1^{(0)} \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \qquad \begin{bmatrix} |5-1| \\ |5-2| \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$
$$\begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \end{bmatrix} = \begin{bmatrix} 5 - 2x_2^{(1)} \\ 5 - 3x_1^{(1)} \end{bmatrix} = \begin{bmatrix} -5 \\ -10 \end{bmatrix}, \qquad \begin{bmatrix} |-5-1| \\ |-10-2| \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$
$$\begin{bmatrix} x_1^{(3)} \\ x_2^{(3)} \end{bmatrix} = \begin{bmatrix} 5 - 2x_2^{(2)} \\ 5 - 3x_1^{(2)} \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \end{bmatrix}, \qquad \begin{bmatrix} |25-1| \\ |20-2| \end{bmatrix} = \begin{bmatrix} 24 \\ 18 \end{bmatrix}$$

In this case the Jacobi Method fails, as the iteration diverges.

I will give the following theorem without proof because we'll take really long time to prove it. So here is a sufficient condition.

**Definition 2.9** The $n \times n$ matrix $A = (a_{ij})$ is **strictly diagonally dominant** if, for each $1 \leq i \leq n, |a_{ii}| > \sum_{j \neq i} |a_{ij}|$.

★ **Convergence theorem !!!** If the $n \times n$ matrix $A$ is strictly diagonally dominant, then for every $b$ and every starting guess, the Jacobi Method or Gauss-Seidal Method applied to $Ax - b = 0$ converges to the unique solution.

**Returning to Example 2.4**

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

is strictly diagonally dominant because $|3| > |1|$ and $|2| > |1|$. Convergence is guaranteed in this case. On the other hand,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

is not strictly diagonally dominant, and no such guarantee exists.

**Note that** strict diagonal dominance is only a sufficient condition. It means that if $A$ is not strictly diagonally dominant, Jacobi or Gauss-Seidal may still convergence but not guaranteed.

**Example 2.21** Determine whether the matrices

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -5 & 2 \\ 1 & 6 & 8 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 8 & 1 \\ 9 & 2 & -2 \end{bmatrix}$$

are strictly diagonally dominant.

The matrix $A$ is diagonally dominant because $|3| > |1| + |-1|$, $|-5| > |2| + |2|$, and $|8| > |1| + |6|$. $B$ is not, because, for example, $|3| > |2| + |6|$ is not true.

**Rearrange the equations to form a strictly diagonally dominant system:** However, if the first and third rows of $B$ are exchanged, then $B$ is strictly diagonally dominant and Jacobi is guaranteed to converge.